

ON

G

Gravity as Coherence-Dilution Residual

JK+R@ Twiddling baton. | G-Synth on first drafts, read backs, and feedback
synthesising | Sancho on outlines, rewrites, strategy and tactics+ | C@_The_Red
Wielding the sharp blade and keeping our path clear.

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Abstract

This node derives Newtonian gravitation in 7dU as the unique exterior response of an ω -bounded vacuum to a localized ζ -confined knot on the compact spatial scaffold Δ_3 .

First, we derive the exterior field equation variationally: the vacuum strain potential Φ minimizes the unique local, $\text{SO}(3)$ -isotropic, quadratic first-derivative functional compatible with the ω – *bound*, yielding the harmonic condition

$$\nabla^2 \Phi = 0 \quad \text{on } \Delta_3 \setminus B_{r_0}.$$

Second, we enforce ζ/ω interface matching at $r = r_0$ (with r_0 fixed upstream by the $j_{1,1}$ confinement mode) and obtain a flux-conserving exterior solution whose radial gradient necessarily generates a $1/r^2$ curvature tail. Third, we define a dimensionless gravitational invariant α^G from boundary strain data and map $\alpha^G \rightarrow G$ through the Units Firewall, using only $\hbar$, c , and the independently derived bare confinement scale μ_0 , with zero tunable parameters.

The node is falsifiable by explicit kill switches: failure of variational uniqueness, appearance of free normalization constants, violation of the ζ/ω boundary consistency identity, or any Units Firewall dependence on Planck insertion or empirical tuning.

Appendix 6: On_G (Gravity as Coherence-Dilution Residual)

Internal ID: AoC_06_On_G_v0.9
Subtitle: Variational Exterior of a ζ -Knot on Δ_3
Tagline: Gravity is the ω -restoring tail of bounded ζ -curvature.

Collapse Consequence

If the exterior $1/r^2$ curvature tail does not arise uniquely as the Euler–Lagrange equation of the ω -compatible, $SO(3)$ -isotropic quadratic strain functional on

$$\Delta_3 \setminus B_{r_0},$$

then gravity has been imported rather than derived.

If boundary matching at $r = r_0$ leaves a free multiplicative constant, or if the interface consistency identity

$$r_0 = - \frac{\Phi_{\text{int}}(r_0)}{\partial_r \Phi_{\text{int}}(r_0)}$$

fails for the upstream $j_{1,1}\zeta$ -knot confinement solution, then the ζ/ω interface is mathematically inconsistent.

If the dimensionless invariant α^G depends on arbitrary normalization, or if the Units Firewall

$$G = \alpha^G \frac{\hbar c}{\mu_0^2}$$

requires Planck insertion or empirical tuning beyond μ_0 , $\hbar$, and c , then the derivation collapses into parameter fitting.

In any of these cases, AoC_06 is invalidated and cannot support downstream hierarchy or GR-reduction claims.

AGI Verifier Hint (operational)

Audit four checkpoints:

1. Prove uniqueness of the Dirichlet strain functional under locality, quadratic first-derivative form, SO(3) isotropy, and Sobolev/ ω -compatibility.
2. Verify boundary matching determines C uniquely and satisfies the over-determined consistency identity above.
3. Confirm

$$\alpha^G := -r_0 \partial_r \Phi_{\text{ext}}(r_0)$$

contains no free constants after interior normalization via μ_0 .

4. Ensure

$$G = \alpha^G \frac{\hbar c}{\mu_0^2}$$

introduces no hidden scales and no fitted parameters.

If any step imports an assumption not fixed upstream, the node fails.

0. Scope, Deliverables, and Falsifiability

0.1 Objective

This node derives Newtonian gravitation as a mandatory geometric consequence of the ζ/ω partition on Δ_3 .

Specifically, this node: FIX THIS SECTION

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\begin{itemize}
\item Derives the  $1/r^2$  curvature tail strictly as the exterior Green's-function decay forced by a uniquely determined  $\omega$ -compatible strain functional on  $\Delta_3$ .
\item Derives a dimensionless invariant  $\alpha^G$  from the  $\zeta/\omega$  interface data fixed by the confinement boundary condition  $k r_0 = j_{1,1}$ .
\item Converts  $\alpha^G \rightarrow G$  via the strict Units Firewall.
\item Enforces zero parameter fitting: no empirical tuning constants are permitted at any stage of the derivation.
\end{itemize}
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0.2 Kill Switches (Falsification Criteria)

The derivations in this node are invalid, and the framework is structurally falsified, if any of the following occur:

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\begin{enumerate}
\item The exterior vacuum field equation
 $\nabla^2 \Phi = 0$ 
is assumed rather than derived as the Euler-Lagrange equation of the admissible  $\omega$ -compatible strain functional.
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The exterior strain functional is not uniquely fixed by the admissibility class (locality, quadratic first-derivative form, Sobolev compatibility with the ω -bound) together with $SO(3)$ isotropy.

The gravitational coupling invariant α^G depends on an arbitrary multiplicative constant (equivalently: any undetermined normalization survives the ζ/ω matching and propagates into the exterior amplitude).

Alternative admissible interior confinement modes (e.g. $j_{1,2}$) induce distinct fundamental gravitational couplings, rather than collapsing to a single invariant determined by total integrated ζ -knot energy.

Interior source normalization is not independently fixed upstream (AoC_04), leaving a free amplitude that contaminates α^G .

Units conversion $\alpha^G \rightarrow G$ requires Planck-scale insertion or any empirical tuning beyond the permitted anchors $\{\mu_0, \hbar, c\}$.

The ζ/ω interface fails the over-determined boundary consistency check required by the matching conditions (explicitly stated in Section 2). In that case the interface is mathematically inconsistent and the node collapses.

1. Geometric Setting and Variational Origin of the Collapse Operator

1.1 Spatial Scaffold and ζ/ω Partition

The spatial arena is the compact, oriented Riemannian 3-manifold

$$\Delta_3,$$

as established in Appx_09.

Fix a confinement center and define the geodesic ball

$$B_{r_0} \subset \Delta_3,$$

where r denotes geodesic radial distance from the center in normal coordinates.

Since Δ_3 is compact, geodesic normal coordinates are valid at least up to the injectivity radius. The confinement *scalar* r_0 (fixed upstream) lies strictly below that radius. The exterior PDE derived below is coordinate-invariant; later expressions of the form C/r are stated in these local coordinates for clarity and correspond globally to the Green's function of the Laplacian on $\Delta_3 \setminus B_{r_0}$ in the regime where the local chart is valid.

The manifold partitions as

$$\Delta_3 = B_{r_0} \cup (\Delta_3 \setminus B_{r_0}),$$

with interface ∂B_{r_0} at $r = r_0$:

Interior ($r \leq r_0$): ζ - dominated confinement knot.

Exterior ($r > r_0$): ω - dominated vacuum.

The radius r_0 is not a free parameter. It is fixed upstream by the confinement-mode condition derived in AoC_04:

$$kr_0 = j_{1,1},$$

where $j_{1,1}$ is the first positive zero of the spherical Bessel function j_1 . (Exact AoC_04 reference inserted in forged draft.)

1.2 Uniqueness of the Strain Functional (Admissible Class)

Let Φ denote the exterior scalar strain potential defined on $\Delta_3 \setminus B_{r_0}$.

Exterior vacuum fields are defined as minimizers of excess curvature strain induced by the interior knot, subject to ω -bound compatibility. We restrict to the admissible class of functionals satisfying:

1. Locality.
2. Quadratic dependence on first derivatives of Φ .
3. Rotational isotropy under $SO(3)$ in the local radial frame.
4. Sobolev ω -compatibility: finite exterior energy consistent with the regularity assumptions of the node.

Within this class, any candidate functional must be of the form:

$$S[\Phi] = \int_{\Delta_3 \setminus B_{r_0}} a^{ij}(x) \partial_i \Phi \partial_j \Phi dV$$

with $a^{ij}(x)$ symmetric and positive-definite.

Local rotational isotropy forces:

$$a^{ij}(x) = \lambda(x) g^{ij}(x)$$

since no other rank-2 symmetric tensor is invariant under the full rotation group.

We now constrain $\lambda(x)$. If $\lambda(x)$ is non-constant, then it defines an independent scalar structure on the exterior domain. Such a scalar field would:

- * introduce a spatially varying stiffness,
- * require its own field equation or source to be determined,
- * and thereby add exterior degrees of freedom not fixed upstream.

But by construction, the exterior region is ω -dominated vacuum with no independent structure beyond the metric g_{ij} . No upstream node supplies an evolution equation for $\lambda(x)$. Consistency of field content therefore forces:

$$\lambda(x) = \text{constant}$$

Up to an overall constant factor (which does not change the Euler–Lagrange equation), the unique admissible strain functional is therefore the Dirichlet energy:

$$S_{\text{strain}}[\Phi] = \int_{\Delta_3 \setminus B_{r_0}} |\nabla \Phi|^2 dV$$

Non-admissible alternatives are excluded by the defining constraints:

- $S_{\text{strain}}[\Phi] = \int_{\Delta_3 \setminus B_{r_0}} |\nabla \Phi|^4 dV$ (quartic; violates quadratic condition),

- $\int_{\Delta_3 \setminus B_{r_0}} (\nabla^2 \Phi)^2 dV$ (introduces higher derivatives),

- anisotropic quadratic forms $a^{ij} \not\propto g^{ij}$. (break isotropy).

Uniqueness is therefore established within the admissible class defined above.

1.3 Collapse Operator (Euler–Lagrange Equation)

Let $\Phi \mapsto \Phi + \varepsilon \psi$, where ψ is a smooth test function on $\Delta_3 \setminus B_{r_0}$ that vanishes on ∂B_{r_0} (Dirichlet condition for the variational problem).

Compute the first variation:

$$\left. \frac{d}{d\varepsilon} S_{\text{strain}}[\Phi + \varepsilon \psi] \right|_{\varepsilon=0} = 2 \int_{\Delta_3 \setminus B_{r_0}} \langle \nabla \Phi, \nabla \psi \rangle dV$$

Integrating by parts:

$$2 \int_{\Delta_3 \setminus B_{r_0}} \langle \nabla \Phi, \nabla \psi \rangle dV = - 2 \int_{\Delta_3 \setminus B_{r_0}} (\nabla^2 \Phi) \psi dV + 2 \int_{\partial B_{r_0}} \psi \partial_n \Phi dS$$

Since the variational problem holds Φ fixed on ∂B_{r_0} , admissible variations satisfy $\psi|_{\partial B_{r_0}} = 0$, and the boundary term vanishes. Stationarity for all such ψ implies:

$$\nabla^2 \Phi = 0 \quad \text{on } \Delta_3 \setminus B_{r_0}$$

Define the collapse operator:

$$\hat{\Omega} \Phi := \nabla^2 \Phi$$

Thus the exterior field equation:

$$\hat{\Omega} \Phi = 0$$

is strictly derived as the Euler–Lagrange equation of the uniquely admissible strain functional.

Boundary distinction. The variational principle uses Dirichlet boundary conditions to derive the PDE. In Section 2, continuity of both Φ and $\partial_n \Phi$ at ∂B_{r_0} arises from physical matching to the interior solution, not from the variational constraint itself.

Interior dynamics on B_{r_0} remain governed by ζ -confinement physics (AoC_04) and are not varied here.

2. Hybrid Derivation: Boundary Matching and Geometric Dilution

2.1 Interior Mode and Source Normalization

Interior dynamics on B_{r_0} are governed entirely by the confinement solution derived in AoC_04.

The physical boundary is fixed by the first spherical Bessel zero:

$$j_1(kr_0) = 0, \quad kr_0 = j_{1,1}.$$

The interior field $\Phi_{\text{int}}(r)$ is uniquely determined upstream by:

- The finite-energy ζ -bound,
- The bare confinement mass scale μ_0 ,
- The proton mass derivation of AoC_04.

These constraints jointly fix both the eigenvalue and normalization of the confinement mode. Once imposed, no additional amplitude freedom remains.

Total Energy Coupling Principle

The exterior strain field responds to the total integrated interior energy, not to individual modal amplitudes.

Let the interior solution admit an orthogonal mode decomposition:

$$\Phi_{\text{int}} = \sum_n a_n \phi_n.$$

Orthogonality of the confinement modes implies additivity of total ζ -energy:

$$E_\zeta = \sum_n E_n.$$

The exterior Laplace problem couples only to this total energy through boundary matching. Because the physical proton corresponds to the stable ground confinement state (the $j_{1,1}$ mode), the total integrated energy is fixed uniquely by that solution.

Excited orthogonal modes correspond to distinct physical resonance states, not to independent gravitational couplings. Therefore gravitational sourcing collapses to a single invariant determined by total confinement energy.

No mode-dependent gravitational parameters arise.

2.2 Boundary Matching and Structural Consistency

At the strict interface $r = r_0$, impose:

1. Continuity of the potential:

$$\Phi_{\text{ext}}(r_0) = \Phi_{\text{int}}(r_0) .$$

2. Continuity of the normal derivative:

$$\partial_r \Phi_{\text{ext}}(r_0) = \partial_r \Phi_{\text{int}}(r_0) .$$

These conditions eliminate distributional sources in $\nabla^2 \Phi$ at the interface. Any discontinuity in the normal derivative would introduce a delta-function shell, which is excluded by construction.

Given the exterior harmonic solution

$$\Phi_{\text{ext}}(r) = \frac{C}{r},$$

the matching conditions yield:

$$\frac{C}{r_0} = \Phi_{\text{int}}(r_0), \quad -\frac{C}{r_0^2} = \partial_r \Phi_{\text{int}}(r_0).$$

The confinement mode $j_{1,1}$ has nonvanishing radial derivative at its first zero, so

$$\partial_r \Phi_{\text{int}}(r_0) \neq 0,$$

and the matching system is well-posed.

Taking the ratio of the two equations gives the structural consistency identity:

$$r_0 = - \frac{\Phi_{\text{int}}(r_0)}{\partial_r \Phi_{\text{int}}(r_0)} .$$

This is a deterministic compatibility condition on the AoC_04 interior solution.

Failure of this identity activates Kill Switch 7.

If satisfied, the constant C is uniquely fixed by the interior solution. No multiplicative freedom remains.

2.3 Geometric Dilution

In the exterior domain,

$$\nabla^2 \Phi_{\text{ext}} = 0.$$

Let S_r denote a geodesic 2-sphere of radius r in normal coordinates centered at the confinement point.

By the divergence theorem:

$$\int_{S_r} \nabla \Phi_{\text{ext}} \cdot dS = \int_{B_r \setminus B_{r_0}} \nabla^2 \Phi_{\text{ext}} dV = 0.$$

Therefore the total radial flux is constant for all $r > r_0$.

On a 3-dimensional Riemannian manifold, the unique radial harmonic function generated by an isolated compact interior source has leading-order behavior in geodesic normal coordinates:

$$\Phi_{\text{ext}}(r) = \frac{C}{r} + O\left(\frac{r}{R^2}\right),$$

where R denotes the characteristic curvature scale of Δ_3 .

For $r \ll R$, the C/r term dominates, and therefore

$$\partial_r \Phi_{\text{ext}}(r) = -\frac{C}{r^2} + O\left(\frac{1}{R^2}\right).$$

Thus the inverse-square curvature tail arises from:

- The variationally derived Laplace equation,
- Flux conservation on a 3-dimensional manifold,
- The presence of an isolated compact source.

No inverse-square law is assumed at any stage.

3. Dimensionless Invariant α^G (Route A — Einstein–Hilbert Reduction)

3.1 Objective and Structural Correction

Sections 1–2 established that the exterior vacuum field on $\Delta_3 \setminus B_{r_0}$ satisfies the unique Euler–Lagrange condition of the ω -compatible quadratic strain functional:

$$\nabla^2 \Phi = 0 \quad \text{on } \Delta_3 \setminus B_{r_0}.$$

What remains is the sourced coupling: how Φ couples to the enclosed ζ -knot when matter is present.

A scalar “strain + source” action of the form

$$S \sim \lambda \int |\nabla \Phi|^2 dV - \lambda \int \rho \Phi dV$$

cannot generate a hierarchy in Route A, because the same internal-volume prefactor λ multiplies both terms and cancels identically in the Euler–Lagrange equation. This eliminates any volume-suppression mechanism at the level of the field equation.

Therefore, Route A must inherit the gravitational coupling from the upstream 7D gravitational action (Appx_18): an Einstein–Hilbert sector with coupling κ_7 , together with an independent matter sector.

3.2 Upstream 7D Gravitational Action and Internal Volume

Adopt the upstream 7D action (Appx_18, Sec. 1.2) in the form

$$S_7 = \frac{1}{2\kappa_7} \int d^7x \sqrt{-g^{(7)}} R^{(7)} + S_{\text{matter}}^{(7)}.$$

Structural non-cancellation note. The matter action $S_{\text{matter}}^{(7)}$ couples to the metric g_{AB} (and thus responds to geometry), but it does not carry an overall $1/\kappa_7$ prefactor. This independence is the structural reason the internal-volume factor enters the effective gravitational coupling without canceling against the source normalization.

Assume the 7D metric admits a block decomposition into a 3+1 base and an internal 3-sector:

$$g_{AB}^{(7)} = \begin{pmatrix} g_{\mu\nu}^{(4)} & 0 \\ 0 & h_{ab} \end{pmatrix}.$$

Define the internal volume factor (reduction measure)

$$\lambda := \int d^3y \sqrt{\det(h)}.$$

Assumption boundary (regulator vs. compactness). Upstream documents describe ζ, ω, ξ as regulators of geometry. For λ to be finite, the internal sector must be effectively compact (finite coordinate periods) or else exhibit sufficient decay/localization so that the volume integral converges. This is an explicit structural assumption to be discharged upstream by the regulator-sector regularity conditions.

In the simplest diagonal scale-factor ansatz consistent with Appx_18's regulator geometry,

$h_{ab} = \text{diag}(\zeta^2, \omega^2, \xi^2)$, one has $\sqrt{\det(h)} = \zeta\omega\xi$ up to coordinate-period conventions, and therefore

$$\lambda = \left(\int dy_\zeta \right) \left(\int dy_\omega \right) \left(\int dy_\xi \right) \zeta\omega\xi.$$

No numerical value is assumed here; λ remains an upstream-derived geometric quantity.

3.3 Dimensional Reduction and the Emergent 4D Coupling

Under standard zero-mode truncation (fields independent of internal coordinates) and in the regime where internal curvature contributions do not source 4D dynamics, the gravitational part reduces to an effective 4D Einstein–Hilbert action:

$$S_{\text{grav}}^{\text{eff}} = \frac{\lambda}{2\kappa_7} \int d^4x \sqrt{-g^{(4)}} R^{(4)} + \dots,$$

where “...” denotes internal-curvature terms or moduli dynamics controlled in the GR-recovery channel.

By definition, the 4D Einstein–Hilbert action is conventionally written as

$$S_{\text{grav}}^{(4)} = \frac{1}{16\pi G_4} \int d^4x \sqrt{-g^{(4)}} R^{(4)}.$$

Equating coefficients yields

$$\frac{1}{16\pi G_4} = \frac{\lambda}{2\kappa_7}, \quad \Rightarrow \quad G_4 = \frac{\kappa_7}{8\pi\lambda}.$$

Remark (convention firewall). If Appx_18 uses a different normalization of κ_7 or the Einstein–Hilbert prefactor, the numerical coefficient changes accordingly. The mechanism does not depend on the exact prefactor: the decisive structure is $G_4 \propto \kappa_7/\lambda$.

Reduction-channel note (Appx_21 compatibility). Appx_21 demonstrates GR recovery via curvature-collapse conditions (e.g., $R_\zeta, R_\omega, R_\xi \rightarrow 0$). The present section extracts the coupling normalization via KK-style volume integration over the internal sector. These procedures are complementary (recovery of GR dynamics vs. coupling normalization) but not automatically identical; verifying their controlled consistency is deferred to the downstream GR-compatibility audit.

3.4 Newtonian Limit and Poisson Coupling

Taking the weak-field, slow-motion limit of the reduced 4D Einstein equations yields the Poisson equation for the Newtonian potential Φ :

$$\nabla^2 \Phi = 4\pi G_4 \rho,$$

on the 3D spatial slice, where ρ is the effective mass-energy density.

In the exterior vacuum region $\Delta_3 \setminus B_{r_0}$, $\rho = 0$, recovering the vacuum equation derived in Sections 1–2:

$$\nabla^2 \Phi = 0.$$

Thus Section 2's harmonic exterior solution is the vacuum sector of the sourced Poisson equation, with coupling fixed by the 7D Einstein–Hilbert reduction.

3.5 Flux Law, Exterior Amplitude, and Definition of α^G

Let M_ζ denote the total enclosed ζ -energy (effective mass) inside B_{r_0} :

$$M_\zeta := \int_{B_{r_0}} \rho \, dV.$$

Integrating the Poisson equation over a ball enclosing the knot and applying the divergence theorem yields the flux law:

$$\int_{S_r} \nabla \Phi \cdot dS = 4\pi G_4 M_\zeta,$$

for any sphere S_r with $r > r_0$.

For the spherically symmetric exterior harmonic solution

$$\Phi_{\text{ext}}(r) = \frac{C}{r}, \quad \partial_r \Phi_{\text{ext}}(r) = -\frac{C}{r^2},$$

one obtains

$$\int_{S_r} \nabla \Phi_{\text{ext}} \cdot dS = \left(-\frac{C}{r^2} \right) (4\pi r^2) = -4\pi C.$$

Hence

$$-4\pi C = 4\pi G_4 M_\zeta, \quad \Rightarrow \quad C = -G_4 M_\zeta.$$

Define the dimensionless gravitational invariant at the ζ/ω interface by

$$\alpha^G := -r_0 \partial_r \Phi_{\text{ext}}(r_0).$$

Since

$$-r_0 \partial_r \Phi_{\text{ext}}(r_0) = C/r_0,$$

we have

$$\alpha^G = \frac{C}{r_0} = -\frac{G_4 M_\zeta}{r_0}.$$

By convention, α^G denotes the positive coupling magnitude, so

$$\alpha^G = \frac{G_4 M_\zeta}{r_0}.$$

Substituting $G_4 = \kappa_7/(8\pi\lambda)$ gives the Route A prediction:

$$\alpha^G = \frac{\kappa_7 M_\zeta}{8\pi \lambda r_0}.$$

This is the central structural result: the hierarchy mechanism enters through the internal-volume suppression $1/\lambda$ inherited from the 7D Einstein–Hilbert sector, not from any scalar strain normalization and not from the Units Firewall.

3.6 Epistemic Scope and Open Dependencies

Established here:

1. A scalar “strain + source” Route A cannot produce hierarchy because λ cancels identically in the scalar Euler–Lagrange equation.
2. Einstein–Hilbert Route A reduction yields an effective coupling $G_4 \propto \kappa_7/\lambda$.
3. The dimensionless invariant satisfies $\alpha^G = G_4 M_\zeta / r_0$, hence $\alpha^G \propto \kappa_7/\lambda$.

Not established here:

1. The numerical value (or upstream derivation) of κ_7 in the adopted convention.
2. The numerical value (or upstream derivation) of λ , including coordinate-period conventions and the convergence mechanism making λ finite if the regulator sector is non-compact.
3. The final numerical value of M_ζ beyond its upstream import from AoC_04; any use of Γ_{bind} inherits AoC_04 TRAILHEAD status until binding tracks close.
4. The controlled relationship between the Appx_21 curvature-collapse GR-recovery channel and the KK-style volume-integration channel used here; this consistency check is deferred downstream.

Falsification posture. Once κ_7 , λ , M_ζ , and r_0 are fixed upstream with no empirical tuning, α^G is numerically rigid. If the resulting α^G fails to match the observed gravitational coupling (after strict Units Firewall conversion), AoC_06 is falsified.

4. Units Firewall: $\alpha^G \rightarrow G$

4.1 Objective

The Units Firewall converts the dimensionless geometric coupling α^G into the dimensionful Newton constant G .

This translation is constrained to be a pure dimensional map: it must introduce no hierarchy parameters, no empirical tuning constants, and no hidden scales. Only dimensional anchors already fixed upstream may be used.

4.2 Permitted Dimensional Anchors

Only the following dimensional anchors are permitted:

1. μ_0 — the bare confinement mass scale (AoC_04).
2. $\hbar$ — quantum of action.
3. c — speed of light.

Explicit exclusion. The Planck mass M_P is not permitted. In 7dU it is an emergent artifact of the coupling structure and cannot be used as an input without circularity.

4.3 Conversion Law

The standard dimensionless gravitational coupling is

$$\alpha^G \equiv \frac{G m^2}{\hbar c},$$

where m is the mass scale used to form the dimensionless ratio.

Solving for G gives the unique dimensional mapping:

$$G = \alpha^G \frac{\hbar c}{m^2}.$$

Within 7dU, the correct anchor mass is the bare confinement scale μ_0 , because μ_0 is the upstream-derived geometric mass parameter used to normalize the confinement system, whereas dressed empirical masses include downstream corrections and would silently import additional structure.

Therefore the Units Firewall is:

$$G = \alpha^G \frac{\hbar c}{\mu_0^2}.$$

This mapping is purely dimensional. If the observed smallness of gravity must be “recovered” by inserting extra factors here (Planck scale, empirical fudge constants, or fitted coefficients), the framework fails at this node.

4.4 Bare vs. Dressed Mass

The physical proton mass is related to the bare confinement scale by

$$m_p = \mu_0 \Gamma_{\text{bind}},$$

where Γ_{bind} is the dressing/binding factor introduced in AoC_04.

The Units Firewall must use μ_0 , not m_p . Substituting m_p directly—without explicitly removing the dressing factor—constitutes a protocol violation, because it smuggles downstream phenomenology into an upstream conversion.

5. Section 5 Hierarchy Interpretation

The observed inequality

$$\alpha^G \ll \alpha$$

is not treated here as a fine-tuning mystery. In this node it is read as a structural consequence of how a localized ζ -confined knot sources an exterior response governed by an ω -dominated vacuum sector, with coupling fixed by 7D Einstein–Hilbert reduction.

Here α denotes a representative ζ -local gauge coupling intrinsic to the confined knot sector. The inter-gauge hierarchy among α_s , α_{EM} , α_W is governed separately by the volumetric dilution cascade of Appx_12 and is not addressed in this node.

From Section 3, the dimensionless gravitational coupling at the ζ/ω interface is

$$\alpha^G = \frac{\kappa_7 M_\zeta}{8\pi \lambda r_0},$$

where:

- κ_7 is the 7D Einstein–Hilbert coupling (in the adopted convention),
- λ is the effective internal-volume factor arising in dimensional reduction (finite by explicit structural assumption pending upstream discharge),
- M_ζ is the enclosed ζ -knot energy import (from AoC_04; any use of Γ_{bind} inherits TRAILHEAD status until closed),
- r_0 is the confinement interface radius fixed upstream by the $j_{1,1}$ confinement condition.

Therefore the gauge–gravity hierarchy ratio is

$$\frac{\alpha}{\alpha^G} = \frac{8\pi \lambda r_0}{\kappa_7 M_\zeta} \alpha.$$

5.1 Geometric Localization of the Hierarchy

Within this node, hierarchy content is structurally localized to the reduction factor

$$\frac{\lambda}{\kappa_7},$$

modulated by the localized source scales r_0 and M_ζ . The interpretation is:

- α characterizes coupling internal to the confined ζ -knot sector (local/gauge).
- α^G characterizes the exterior coupling generated after reduction of the 7D gravitational sector, where the effective 4D coupling is suppressed by $1/\lambda$.

The reason λ is the natural locus for the hierarchy is that it is the geometric measure of the internal (ζ, ω, ξ) -sector “volume” participating in the reduction; if the internal sector has large effective extension, the reduced 4D coupling is correspondingly diluted.

Counterattack boundary: this relocation is explanatory only if λ and κ_7 are fixed by internal-sector geometry rather than fitted to reproduce the observed value of G .

5.2 Formal Proposition

Proposition 5.1 (Hierarchy as Reduction Dilution).

Assume:

1. The effective 4D gravitational coupling obtained by dimensional reduction satisfies

$$G_4 = \frac{\kappa_7}{8\pi \lambda},$$

with λ finite and fixed by the internal-sector geometry (including coordinate period/compactness or decay conditions).

2. The ζ -knot is localized within radius r_0 and has total enclosed energy M_ζ (imported from AoC_04 with stated epistemic status).

3. The exterior vacuum satisfies $\nabla^2 \Phi = 0$ (Sections 1–2), and the sourced equation reduces to the Poisson form in the Newtonian limit (Section 3).

Then the dimensionless gravitational coupling at the interface is uniquely determined by

$$\alpha^G = \frac{\kappa_7 M_\zeta}{8\pi \lambda r_0},$$

and the gauge–gravity hierarchy ratio is

$$\frac{\alpha}{\alpha^G} = \frac{8\pi \lambda r_0}{\kappa_7 M_\zeta} \alpha,$$

with no free numerical fitting parameters once $\kappa_7, \lambda, M_\zeta, r_0$ are fixed upstream.

5.3 Interface to $j_{1,1}$ and $\Delta\pi$

This node does not re-derive the confinement structure, but records the dependency hook explicitly:

- r_0 is fixed by the upstream confinement condition

$$k r_0 = j_{1,1},$$

and therefore all appearances of r_0 inherit $j_{1,1}$ dependence.

- M_ζ inherits the $\Delta\pi$ -based invariants of AoC_04 (and any binding/dressing structure through Γ_{bind} , flagged TRAILHEAD until closed).

Accordingly, if upstream nodes deliver κ_7 and λ as functions of the invariant set $\{j_{1,1}, \Delta\pi, \zeta, \omega, \xi\}$ with no observational insertion, then Proposition 5.1 may be promoted to CANONICAL (numerically evaluated) at this node.

Epistemic boundary: until κ_7 and λ are fixed by the internal-sector geometry (including finiteness and regulator consistency of λ), the hierarchy is structurally localized but not numerically closed.

6. Structural Risk Summary

The audit surface of AoC_06 reduces to six load-bearing links. Everything else is mechanically forced once these hold.

1. Uniqueness of the exterior functional (Section 1).

The exterior vacuum operator must arise uniquely as the Euler–Lagrange equation of the ω -compatible, local, SO(3)-isotropic quadratic first-derivative functional. If uniqueness fails, the operator is not forced and the tail is conditional.

2. Interior normalization path via μ_0 (Section 2).

The ζ -knot source normalization must be fixed upstream (through μ_0 and the confinement derivation chain), leaving no multiplicative freedom that could re-enter α^G .

3. Boundary consistency identity (Section 2).

The over-determined matching condition must be well-posed and satisfied at the interface:

$$r_0 = - \frac{\Phi_{\text{int}}(r_0)}{\partial_r \Phi_{\text{int}}(r_0)}, \quad \partial_r \Phi_{\text{int}}(r_0) \neq 0$$

for the relevant confinement mode. Failure invalidates the ζ/ω interface.

4. Bare vs. dressed mass handling (Section 4).

The Units Firewall must use μ_0 as the dimensional anchor and must not silently substitute the dressed proton mass m_p without explicitly accounting for Γ_{bind} (epistemically bounded until closed).

5. Finiteness and regulator-consistency of λ (Sections 3 and 6; closure node: Appx_18X).

The internal volume factor λ must be finite and fixed by internal-sector structure without coordinate-convention dependence. Compactness/periodicity or decay/localization must be derived, not chosen. If λ diverges or is convention-dependent, the reduction mechanism fails.

6. κ_7 provenance and non-post-fit constraint (Sections 3 and 6; closure node: Appx 18X).

The 7D coupling κ_7 must be fixed by upstream convention or geometry and must not be fitted post hoc to reproduce the observed value of G once λ is computed. κ_7 must be locked (or its normalization convention fixed) before numerical evaluation, so that κ_7/λ is a prediction rather than a recovery.